

SQL Date Functions Exercise

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Question 1

SELECT

Order-Id,

Customer-Id,

Order-date,

Dayname (Order-date) As day-name

From orders;

Order-Id	Customer-Id	Order-date	day-name
1001	100	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

Question 2

SELECT

Customer-Id,

Customer-Name,

Signup-date,

Monthname (Signup-date) As Signup-month

From Customer-Signups;

Customer-Id	Customer-Name	Signup-date	Signup-m
201	John	2026-01-15	January
202	Mary	2026-02-20	February
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May

Question 3

SELECT

Sale-Id,

Product-Name,

Sale-date,

Month(Sale-date) AS Sale-month

FROM sales;

Sale-Id	Product-Name	Sale-date	Sale-month
S001	Laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	Keyboard	2026-03-20	3
S004	Monitor	2026-04-25	4
S005	Desk	2026-05-30	5

Question 4

SELECT

transaction-Id,

Customer-Id,

transaction-date,

YEAR(transaction-date) AS transaction-year

FROM transactions;

transaction-Id	Customer-Id	transaction-date	transaction-year
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

Question 5

SELECT

delivery-Id,

Customer-Id,

delivery-date,

DAY(delivery-date) AS day-of-month

FROM deliveries ;

delivery-id	Customer-id	delivery-date	day of month
D001	103	2026-05-01	1
D002	103	2026-05-08	8
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

Question 6

SELECT

employee-id,
employee-name,
department,

CURRENT-DATE() AS today-date

FROM employees;

employee-id	employee-name	department	today-date
301	Nandi	Sales	2026-08-01
302	Brian	IT	2026-08-01
303	Lerato	Finance	2026-08-01
304	Sipho	HR	2026-08-01
305	Aisha	Marketing	2026-08-01

Question 7

SELECT

order-id,

customer-id,

order-date-text,

TO-DATE(order-date-text) AS order-date

FROM online-orders;

order-id	Customer-id	order-date-text	order-date
4001	101	2026-01-15	2026-01-15
4002	102	2026-02-20	2026-02-20
4003	103	2026-03-25	2026-03-25
4004	104	2026-04-10	2026-04-10
4005	105	2026-05-05	2026-05-05

QUESTION 8

SELECT

Payment-id,

Customer-id,

Payment-date,

TO CHAR (Payment-date, 'YYYY-MM-DD')

AS Formatted-Payment-date

FROM payment-dates;

Payment-id	Customer-id	Payment-date	Formatted-P-date
P001	101	2026-01-05	2026-01-05
P002	102	2026-02-10	2026-02-10
P003	103	2026-03-15	2026-03-15
P004	104	2026-04-20	2026-04-20
P005	105	2026-05-25	2026-05-25

Question 9

SELECT

Customer-id,

Customer-name,

last-purchase-date,

DATEDIFF (CURRENT-DATE(), last-purchase-date)

AS days-since-last-purchase

FROM customer-purchase;

customer - id	customer - name	last - purchase - date	days - since last purchase
501	John	2026-05-01	92
502	Mary	2026-05-10	83
503	Peter	2026-05-15	78
504	Sarah	2026-05-20	73
505	Thabo	2026-05-25	68

Question 10

SELECT

order - id,

customer - id,

order - date,

DATEADD (DAY, 7, order - date) AS expected - delivery - date

FROM Shipping - Orders;

order - id	customer - id	order - date	expected - del - date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

Question 11

SELECT

booking - id,

customer - id,

booking - date,

YEAR (booking - date) AS booking - year,

MONTH (booking - date) AS booking - month,

DAY (booking - date) AS booking - day

FROM bookings

booking-id booking-id	customer-id customer-id	booking-date booking-date	booking-year booking-year	booking-month booking-month	booking-day booking-day
Boo1	101	2026-01-12	2026	1	12
Boo2	102	2026-02-18	2026	2	18
Boo3	103	2026-03-22	2026	3	22
Boo4	104	2026-04-09	2026	4	9
Boo5	105	2026-05-27	2026	5	27

Question 12

SELECT

Order-id,

Customer-id,

Order-date,

YEAR(Order-date) AS order-year,

amount

FROM yearly-orders

WHERE YEAR(Order-date) = 2026;

order-id	Customer-id	order-date	order-year	Amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000

Question 13

SELECT

order-id

Customer-id

Order-date,

Month(Order-date) AS order-month,

amount

FROM monthly-orders

WHERE MONTH(Order-date) = 3;

DATE	Month
2026-01-15	1
2026-02-20	2
2026-03-10	3
2026-03-25	3
2026-05-30	5

Question 14

SELECT

Subscription - Id,

Customer - Id,

Start-date,

LAST-DAY(Start-date) AS Month-end-date,
FROM Subscription;

Start Date	LAST DAY
2026-01-10	2026-01-31
2026-02-15	2026-02-28
2026-03-20	2026-03-31

Question 15

SELECT

Send - Id,

Customer - Id,

Send-date,

DATE - TRUNC('MONTH', send-date)

AS Month-start-date

FROM Campaign - Sends;

Send - Id	Customer - Id	Send-date	Month-start-date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-09	2026-04-01
C005	105	2026-05-27	2026-05-01

QUESTION 16

SELECT

INVOICE-ID,

CUSTOMER-ID,

INVOICE-DATE,

TO-CHAR(INVOICE-DATE, 'Month YYYY')

AS INVOICE-MONTH-YEAR

FROM INVOICE-DATES;

INVOICE-ID	CUSTOMER-ID	INVOICE-DATE	INVOICE-MONTH-YEAR
INV001	101	2026-01-05	Jan 2026
INV002	102	2026-02-10	Feb 2026
INV003	103	2026-03-15	Mar 2026
INV004	104	2026-04-20	Apr 2026
INV005	105	2026-05-25	May 2026

QUESTION 17

SELECT;

CUSTOMER-ID,

CUSTOMER-NAME,

DATE-OF-BIRTH,

YEAR(CURRENT-DATE()) - YEAR(DATE-OF-BIRTH)

AS CUSTOMER-AGE

FROM CUSTOMER-BIRTHDAYS;

CUSTOMER-ID	CUSTOMER-NAME	DATE-OF-BIRTH	CUSTOMER-AGE
901	John	1998-05-10	28
902	Mary	1990-08-20	36
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	41
905	Thabo	2000-07-30	26

Question 18

SELECT

order-id,

Customer-id,

Order-date,

DAYNAME (order-date) AS day-name

CASE

WHEN DAYNAME (order-date) IN ('Saturday',
Sunday)

THEN 'weekend'

ELSE 'weekday'

END AS day-type

FROM weekend-orders;

order-id	Customer-id	order-date	day-name	day-type
9001	101	2026-05-01	Friday	weekday
9002	102	2026-05-02	Saturday	weekend
9003	103	2026-05-03	Sunday	weekend
9004	104	2026-05-04	Monday	weekday
9005	105	2026-05-05	Tuesday	weekday

Question 19

SELECT

transaction-id,

Customer-id,

transaction-date,

Quarter (transaction-date) AS transaction-quarters,

amount

FROM quarterly-transactions;

transaction-id	Customer-id	transaction-date	transaction - quarter	Amount
Q001	101	2026-01-15	1	500
Q002	102	2026-03-20	1	1200
Q003	103	2026-04-10	2	800
Q004	104	2026-07-05	3	1500
Q005	105	2026-10-25	4	2000

Question 20

SELECT

order-id,

customer-id,

order-date,

DATEDIFF(CURRENT-DATE(), order-date)

AS days-since-order,

amount

FROM recent-orders

WHERE DATEDIFF(CURRENT-DATE(), order-date) > 30;

order-id	customer-id	order-date	days-since-order	Amount
R001	101	2026-04-01	122	500
R002	102	2026-04-15	108	1200
R003	103	2026-04-01	92	800
R004	104	2026-05-10	83	1500
R005	105	2026-05-25	68	2000

Bonus Challenge

SELECT customer-id,

customer-name,

last-purchase-date,

DATEDIFF(CURRENT-DATE(), last-purchase-date)

AS days-since-last-purchase,

CASE

WHEN DATEDIFF(CURRENT-DATE(), last-purchase-date

<= 30 THEN 'Active Customer'

WHEN DATEDIFF(CURRENT-DATE(), last-purchase-date)
 BETWEEN 31 AND 90 THEN 'At Risk Customer'
 ELSE 'Inactive customer'

END AS Customer-Status

FROM Customer-recency;

Customer Id	Customer Name	last-purchase date	days-since last-pur	Customer Status
101	John	2026-05-28	68	At Risk Cust
102	Mary	2026-05-10	83	At Risk Cust
103	Peter	2026-04-01	122	Inactive Cust
104	Sarah	2026-02-15	167	Inactive Cust
105	Thabo	2026-12-20	224	Inactive Cust